



Contact

Phone: +33 620 383 526

Email:
stephanie.mattab@gmail.com

LinkedIn:
<https://www.linkedin.com/in/stephanie-mattab/>

GitHub:
<https://github.com/StephanieMattaB>

Projects:
<https://stephaniemattab.github.io/>

Language

- Spanish - Native
- English - Bilingual
- French – B2
- Mandarin - HSK3

Stephanie Matta Bobadilla

Control Engineer & Power Systems Specialist

PRESENTATION

Systems-level engineer specializing in optimization and modern energy systems. Skilled in developing operational decision support for decentralized networks and markets. Currently completing a Triple Degree Master's in dynamics & control of renewable-based power systems.

EXPERTISE

- Computational: Python, Julia, MATLAB, surrogate modeling, multi-agent systems, data pipelines, forecasting, dynamic programming.
- Grid Simulation: MATLAB/Simulink, PSSE, PowerFactory, PSCAD, MATPOWER, GridCal.
- Power System Analysis: stability studies, optimal power flow, EMT simulation, converter-based systems, physics-based constraints.
- Control & Optimization: Algorithm foundations, QP, LP/MILP, stochastic optimization, Generalized Nash Equilibria (GNE), MPC, multi-parametric/combinatorics solutions.
- SciComp: planning and coordination solver development, Git (PRs, code review), CI/CD, scalable scientific software.

Open-source Contributions:

[mpGNE-games](#): Simultaneous optimization with shared physical constraints: A Julia solver for computing how multiple interacting systems will share limited resources, as an explicit piecewise-affine map for any given input.

WORK EXPERIENCE

KTH Royal Institute of Technology

Researcher, Decisions & Control Systems Department

2026-Present

- Designed game-theoretic optimization frameworks (GNE/mpVI) for flexibility coordination in active distribution grids. Developed open-source Julia solvers for fair resource allocation and explicit decision-making across decentralized energy communities within Prof. Giuseppe Belgioioso's group.

SUJIO

R&D Developer & Commercial Engineer

2024-2026

- Built SCADA-EMS data architecture and LSTM forecasting pipeline. Deployed trading decision support reducing forecast error and enabling 10+ GWh/month market bidding strategies with 40% faster decision cycles.
- Founded innovation hub launching 3 R&D initiatives: digital asset data platform, ML energy forecasting, market trading optimization.

ENEL Green Power

2023

Regulatory & Institutional Affairs Intern

- Accelerated grid codes and storage integration policies for hybrid renewable systems, supporting regulatory strategy and system-level planning; co-authored corporate generation mix analytics report and contributed to the first IEA LatAm Energy Outlook.

British Petroleum

2022-2023

EHS & Sustainability Intern

- Managed downstream operational compliance for 50+ audited fuel-retail-sites. Developed solar PV integration techno-economic feasibility studies achieving 30% grid electricity reduction per site.

EDUCATION

M.Sc. Dynamics and Control of Renewables-Based Power Systems · Triple Degree | 2024-2026

École Centrale de Nantes (FR), Polytechnic University of Catalonia (ES), Politehnica University of Bucharest (RO) | Erasmus Mundus Master

Top 2% scholarship awardee.

Relevant Coursework: **power electronics, robust control, stability & optimization.**

- RTE (French Transmission System Operator) Collaboration for Unit Commitment optimization coding (AMPL) in 100+ generation units in renewable-heavy dispatch scenarios. Delivered dispatch planning improvements supporting TSO operations.

B.Sc. Energy & Sustainable Development Engineering (GPA 9.6/10.0) | 2020-2024

Monterrey Institute of Technology · Semester abroad: Tiangong University, China, School of Artificial Intelligence & Control Science.

Relevant Coursework: Large-scale energy projects, thermal and process engineering, industrial and corporate sustainability strategies, industrial business, financing & regulation.

- Independent research collaboration with Beijing JiaoTong University on HVDC systems, simulating commutation faults in PSCAD and training Neural Networks-based fault detection models.

Intensive Technical Programmes: Imperial College London & Tsinghua (Aug–Sept 2025, inverter-based resources & data driven methods, Beijing) · NTUA (Apr 2026, Hardware Validation | Intelligent and Secure Microgrids, Management Systems, Athens).